On Nonlinear Partial Differential Equations with Variable Nonlinearity

Doctoral Day — 15 October 2026

Mahdi Tahar Brahimi
Supervisor: Prof. Abdelhamid Tallab
Co-Supervisor: Dr. Abdelaziz Hellal

Université Mohamed Boudiaf – M'sila
Faculty of Mathematics and Computer Science
Speciality: Functional Analysis for PDEs

15 October 2026



Table of Contents

1. Thesis Overview
2. Year 1 — Foundations
3. Year 2 (2025/2026) — Theoretical Framework
4. Year 3 — Advanced Results & Numerical Implementation
5. Mathematical Foundations
6. Conclusion

Thesis Identity

Title

On Nonlinear Partial Differential Equations with Variable Nonlinearity

French Title

Sur les équations aux dérivées partielles non linéaires avec non-linéarité variable

Field: Mathematics
Speciality: Functional Analysis for PDEs

Supervisor: Prof. A. Tallab
Co-Supervisor: Dr. A. Hellal

Keywords

- Nonlinear elliptic equations
- Nonlinear parabolic equations
- Variable nonlinearity
- Existence, regularity, and uniqueness of solutions
- Variational and non-variational methods
- Sobolev and Lebesgue spaces with variable exponents

Scientific Summary

Research Focus

Theoretical and numerical analysis of nonlinear elliptic and parabolic PDEs characterized by **variable nonlinearity**.

Core Framework

Study of existence, uniqueness, and regularity of solutions within **Sobolev and Lebesgue spaces with variable exponents**.

Goal

Develop new analytical tools and stable numerical methods for this complex class of PDEs, with applications in physics, biology, and engineering.

Year 1: Objectives

1. Develop rigorous mathematical theories for existence, uniqueness, and regularity using functional analysis tools.
2. Design stable and efficient numerical algorithms tailored to these PDEs.
3. Apply theoretical and numerical results to real-world problems in applied sciences.
4. Contribute to interdisciplinary advancements bridging fundamental mathematics with technological challenges.

Year 1: Work Plan

1. **Literature Review:** Critical analysis of existing works on nonlinear PDEs and variable nonlinearity.
2. **Theoretical Analysis:** Proofs of existence, uniqueness, and regularity using variational methods.
3. **Numerical Development:** Implementation of algorithms (finite elements, finite differences).
4. **Applied Validation:** Case studies on concrete problems.
5. **Dissemination:** Publication of results and final thesis writing.

Year 1: Doctoral Training Programme

Activity	**Semester 1**	**Semester 2**
Speciality courses	Sobolev/Lebesgue spaces with variable exponents (3h)	Research methodology (1.5h)

Training Objectives

- Establish rigorous mathematical theories for nonlinear elliptic and parabolic equations.
- Create efficient and stable numerical algorithms.
- Apply results to real-world problems in physics, biology, and engineering.

Year 2: Progress Overview

Overall Progress: Approximately 25%

- ✅ Completed Foundational coursework on variable exponent spaces and research methodology.
- ✅ Completed Thorough literature review and precise formulation of the specific class of PDEs.
- ⏳ In Progress Active development of the theoretical framework; initial a priori estimates underway.
- 🕒 Planned Rigorous numerical analysis and implementation of algorithms.
- 🕒 Planned Detailed application to real-world models (e.g., electrorheological fluids).

Year 2: Work Plan — Months 1–6

Primary Objective

Solidify the theoretical foundation and initiate the numerical analysis phase.

Theoretical Finalization

1. Finalize the proof of the main existence and uniqueness theorem for elliptic and parabolic equations with $p(x)$-growth.
2. Initiate a detailed study of solution regularity (e.g., Hölder continuity, higher integrability).
3. Complete a draft of the first research paper for submission to a peer-reviewed journal.

Year 2: Work Plan — Months 7–12

Bridge to Numerical Methods

1. Begin numerical analysis: stability and convergence of a Finite Element Method adapted to variable exponents.
2. Develop and test a basic prototype algorithm (MATLAB, Python with FEniCS).
3. Produce a working prototype code and a draft section of the thesis on the numerical scheme.

Expected Deliverables

- Working prototype code
- Draft thesis section on numerical scheme
- First research paper draft

Year 3: Summary of Previous Year

Main Achievements

- Completed comprehensive literature review on nonlinear PDEs with variable nonlinearity.
- Studied variational and non-variational methods for existence and uniqueness.
- Began investigating regularity results under variable exponent conditions.
- Attended doctoral training courses (3h + 1.5h).
- Submitted the first paper of the thesis manuscript [2].
- Drafted the first part of the thesis manuscript.

Year 3: Progress of the Work

Component	Status
Theoretical framework	✅ Completed — Functional setting established.
Existence & uniqueness	⏳ In Progress — Elliptic case preliminary; parabolic under study.
Regularity	⏳ In Progress — Preliminary investigation begun.
Numerical methods	🕒 Planned — Planned for current year.
Manuscript	⏳ In Progress — First chapter drafted.

Year 3: Progress Rate

Overall Progress: Approximately 45%

The work is broadly proceeding as planned. A slight delay was caused by the additional time required to master the variable exponent functional framework, which is essential for the rest of the thesis. The revised schedule remains on track for completion within the planned timeframe.

Year 3: Work Plan — Main Objectives

1. Finalise existence and uniqueness results for the elliptic case and extend them to the parabolic case.
2. Establish regularity results for solutions under variable nonlinearity.
3. Begin developing numerical schemes for approximating solutions, with a focus on stability and accuracy.
4. Prepare and submit a first research article to a peer-reviewed journal.
5. Continue drafting the thesis manuscript (Chapters 2 and 3).

Year 3: Planned Activities

Period	Activity
Semester 1	Complete existence, uniqueness proofs; begin regularity analysis; draft article 1.
Semester 2	Extend results to parabolic case; begin numerical methods; draft Chapters 2–3.
Throughout	Regular meetings with supervisors; attend seminars and doctoral training.

New Orientations

- Greater emphasis on numerical approximation of solutions.
- Consideration of applications in physics, biology, and engineering.

Core Definitions: Variable Exponent Spaces [5]

Variable Exponent

$p : \Omega \to [1, \infty)$ measurable

Bounds: $p^- = \operatorname{ess\,inf} p(x)$, $p^+ = \operatorname{ess\,sup} p(x)$

Modular: $\rho_{p(\cdot)}(f) = \int_\Omega |f(x)|^{p(x)} dx$

Lebesgue Space $L^{p(\cdot)}$

$L^{p(\cdot)}(\Omega) = \left\{ f : \exists\, \lambda > 0,\ \rho_{p(\cdot)}(f/\lambda) < \infty \right\}$

Luxemburg Norm:

$\|f\|_{p(\cdot)} = \inf \left\{ \lambda > 0 : \rho_{p(\cdot)}(f/\lambda) \leq 1 \right\}$

Sobolev Space $W^{k,p(\cdot)}$

$W^{k,p(\cdot)}(\Omega) = \left\{ f \in L^{p(\cdot)} : D^\alpha f \in L^{p(\cdot)},\ |\alpha| \leq k \right\}$

Key Properties

- **Banach:** Complete under Luxemburg norm
- **Reflexive:** When $1 < p^- \leq p^+ < \infty$
- **Separable:** When $p^+ < \infty$
- **Hölder:** $\|fg\|_{s(\cdot)} \leq 2\|f\|_{p(\cdot)}\|g\|_{q(\cdot)}$

Essential Conditions [5]

Log-Hölder Continuity

$$|p(x) - p(y)| \leq \frac{C}{-\log|x-y|}, \quad \forall\, x, y \in \Omega,\ 0 < |x-y| < \tfrac{1}{2}$$

Sobolev Embedding

If p is log-Hölder and $1 < p^- \leq p^+ < n$:

$$W^{1,p(\cdot)} \hookrightarrow L^{p^*(\cdot)}, \qquad \frac{1}{p^*(x)} = \frac{1}{p(x)} - \frac{1}{n}$$

Why These Conditions Matter

Log-Hölder continuity ensures the **Poincaré inequality**, **embedding theorems**, and **density of smooth functions** in $W^{1,p(\cdot)}$ — all essential for PDE analysis.

Example 1: Elliptic Equations — $p(x)$-Laplace [7] I

Model Problem

$$-\Delta_{p(x)} u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega$$

where $\Delta_{p(x)} u = \operatorname{div}\left(|\nabla u|^{p(x)-2}\nabla u\right)$

Weak Formulation

Find $u \in W_0^{1,p(\cdot)}(\Omega)$ such that:

$$\int_\Omega |\nabla u|^{p(x)-2}\nabla u \cdot \nabla v \, dx = \int_\Omega fv \, dx$$

for all $v \in W_0^{1,p(\cdot)}(\Omega)$

Energy Functional

$$J(u) = \int_\Omega \frac{1}{p(x)} |\nabla u|^{p(x)} dx - \int_\Omega fu \, dx$$

Minimizers of J are weak solutions.

Example 1: Elliptic Equations — $p(x)$-Laplace [7] II

Existence & Regularity

- **Existence:** If p log-Hölder, $1 < p^- \leq p^+ < \infty$, $f \in L^{p'(\cdot)}(\Omega)$
- **Regularity:** Additional conditions yield $u \in C^{1,\alpha}(\Omega)$ [5]
- **Uniqueness:** Guaranteed if $p(x) \geq 2$ or $1 < p(x) \leq 2$ for all x

Example 2: Parabolic Equations — $p(x)$-Heat I

Model Problem

$$u_t - \Delta_{p(x)} u = f(x,t) \quad \text{in } \Omega \times (0,T)$$

$$u = 0 \quad \text{on } \partial\Omega \times (0,T), \qquad u(x,0) = u_0(x) \quad \text{in } \Omega$$

Weak Formulation

Find $u \in L^2(0,T; W_0^{1,p(\cdot)}(\Omega))$ with $u_t \in L^2(0,T; W^{-1,p'(\cdot)}(\Omega))$ such that:

$$\langle u_t, v \rangle + \int_\Omega |\nabla u|^{p(x)-2} \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx$$

for all $v \in W_0^{1,p(\cdot)}(\Omega)$, a.e. $t \in (0,T)$

Example 2: Parabolic Equations — $p(x)$-Heat II

Energy Estimates

Basic Energy:

$$\frac{1}{2}\frac{d}{dt}\|u\|_2^2 + \int_\Omega |\nabla u|^{p(x)} dx \leq \int_\Omega fu\, dx$$

Higher Integrability: $|\nabla u|^{p(x)} \in L^{1+\delta}(\Omega \times (0,T))$ for some $\delta > 0$.

Variable Exponent Spaces — Formal Framework [5]

Definition 1 (Variable Exponent Lebesgue Space)

For a measurable function $p : \Omega \to [1, \infty)$:

$$L^{p(x)}(\Omega) = \left\{ u : \Omega \to \mathbb{R} \;\middle|\; \int_\Omega |u(x)|^{p(x)}\, dx < \infty \right\}$$

Definition 2 (Variable Exponent Sobolev Space)

$$W^{1,p(x)}(\Omega) = \left\{ u \in L^{p(x)}(\Omega) \;\middle|\; \nabla u \in L^{p(x)}(\Omega) \right\}$$

Poincaré Inequality

$$\|u\|_{p(\cdot)} \leq C \|\nabla u\|_{p(\cdot)}, \quad \forall u \in W_0^{1,p(x)}(\Omega)$$

Main Existence and Uniqueness Theorem — Part 1 [2]

Dirichlet Problem

$$-\operatorname{div}\left(|\nabla u|^{p(x)-2}\nabla u\right) = \beta|\nabla u|^{q(x)} + f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega$$

with $\Omega \subset \mathbb{R}^N$, $N \geq 2$, $p, q : \overline{\Omega} \to (1, N)$, $p(x) - 1 < q(x) < p(x)$.

Existence

If $f \in L^{m(\cdot)}(\Omega)$ with suitable summability, then

$$u \in W_0^{1,p(x)}(\Omega)$$

Uniqueness

Let $\frac{2N}{N+2} \leq p(x) \leq 2$ and

$$p(x) - 1 < q(x) \leq p(x) - 1 + \frac{p(x)}{N}.$$

Then the solution is **unique**.

Main Existence and Uniqueness Theorem — Part 2 [1]

Comparison Principle

If $f \leq g$ in $\mathcal{D}'(\Omega)$, then $u \leq v$ a.e. in Ω.

Three Conditions for q

1. $p - 1 < q < \dfrac{N(p-1)}{N-1}$ (Renormalized/Entropy solutions)
2. $\dfrac{N(p-1)}{N-1} \leq q < p - 1 + \dfrac{p}{N}$ (Weak solutions)
3. $p - 1 + \dfrac{p}{N} \leq q \leq p$ (Standard weak solutions)

Sharp Smallness Condition

If $\|f\|_{L^{(p^*)'}} < K_2$ and $\|g\|_{L^{(p^*)'}} < K_2$, then $u \leq v$ a.e., where

$$K_2 = \frac{q-p+1}{q} \left(\omega_N^{1/N} \frac{N(q-p+1)-q}{q-p+1} \right)^{\frac{q-p+1}{q}} \left(\frac{p-1}{\beta q} \right)^{\frac{N(q-1)}{N(q-p+1)}}$$

Key Theorems [5, 4]

Theorem 3 (Existence and Uniqueness)

Under appropriate conditions on the variable exponent $p(x)$ and the nonlinearity, the elliptic problem

$$-\operatorname{div}\left(|\nabla u|^{p(x)-2}\nabla u\right) = f(x, u, \nabla u)$$

admits a unique weak solution $u \in W_0^{1,p(x)}(\Omega)$.

Theorem 4 (Regularity)

If $p(x)$ is log-Hölder continuous and f satisfies suitable growth conditions, then the weak solution is locally Hölder continuous [5].

Key Tool: Three-Step Method [3]

Approximation $\rightarrow$ A Priori Estimates $\rightarrow$ Passage to the Limit.

Numerical Scheme: Finite Element Method [4, 6]

Discrete Problem

Find $u_h \in V_h \subset W_0^{1,p(x)}(\Omega)$ such that for all $v_h \in V_h$:

$$\int_\Omega |\nabla u_h|^{p(x)-2} \nabla u_h \cdot \nabla v_h \, dx = \int_\Omega f \, v_h \, dx$$

Key Challenges

- Variable exponent affects stiffness matrix assembly.
- Nonlinearity requires iterative solvers (Newton, fixed-point).
- Stability and convergence analysis in variable exponent spaces.

Physical Applications [9, 7]

Electrorheological Fluids [9]

$p(x)$ depends on the electric field — a classic application of variable exponent PDEs.

Image Processing

Adaptive regularization via variable exponent models.

Non-Newtonian Fluids

Shear-dependent viscosity modeled by $p(x)$-Laplacian.

Other Contexts

- Thermal processes with temperature-dependent conductivity
- Filtration in porous media
- Biological systems with density-dependent diffusion

Summary of Progress

Year 1	Year 2	Year 3
Foundations	**Theory**	**Implementation**
Literature review	Existence proofs	Numerical schemes
Coursework	Regularity	Applications
Problem formulation	Numerical bridge	Manuscript

Current Status: 45% Complete

On track for completion within the planned timeframe.

References I

- Alvino, A., Ferone, V., & Mercaldo, A. (2025). The Existence and Uniqueness of Nonlinear Elliptic Equations with General Growth in the Gradient. *Mathematics*, 13, 63.
- Brahimi, M.-T., & Hellal, A. (2026). Dirichlet boundary value problems with general growth in the gradient involving variable exponents. *Preprint*.
- Boccardo, L., & Murat, F. (1992). Almost everywhere convergence of the gradients of solutions to elliptic and parabolic equations. *Nonlinear Analysis*, 19, 581–597.
- Chipot, M. (2000). *Elements of Nonlinear Analysis*. Birkhäuser.
- Diening, L., Harjulehto, P., Hästö, P., & Růžička, M. (2011). *Lebesgue and Sobolev Spaces with Variable Exponents*. Springer.
- Evans, L. C. (2010). *Partial Differential Equations* (2nd ed.). AMS.

References II


Fan, X. (2015). An overview of $p(x)$-Laplace equations. *Nonlinear Analysis*.


Lions, J.-L. (1969). *Quelques méthodes de résolution des problèmes aux limites non linéaires*. Dunod.


Růžička, M. (2000). *Electrorheological Fluids: Modeling and Mathematical Theory*. Springer.


Showalter, R. E. (1997). *Monotone Operators in Banach Space and Nonlinear Partial Differential Equations*. AMS.

Thank You!

Questions?

Mahdi Tahar Brahimi
Université Mohamed Boudiaf – M'sila
Speciality: Functional Analysis for PDEs

Contact: mahdithar.brahimi@univ-msila.dz